

LAB:

Python CRUD with MySQL (Complete Real-Time Project)

We'll build a **simple Employee Management System**.

Operations:

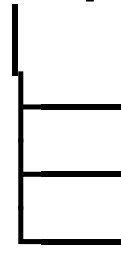
- . Create Employee
- . Read Employees
- . Update Employee
- . Delete Employee

Technology:

- . Python
- . MySQL
- . VS Code

Project Structure

employee_crud/



- db.py
- employee_crud.py
- requirements.txt

I U D

Step 1: Install Python package

Open terminal.

pip install mysql-connector-python

Create requirements.txt

mysql-connector-python

File Edit Selection View Go Run ...

0employee_crud

EXPLORER

0EMPLOYEE_CRUD

db.py

employee_crud.py

requirements.txt

db.py

employee_crud.py

1

Generate code (Ctrl+I), or select a language (Ctrl+K M). Start

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

C:\-----Working-----\0_Hyd_Vardamaan\0employee_crud>pip install mysql-c

'pip' is not recognized as an internal or external command,

operable program or batch file.

C:\-----Working-----\0_Hyd_Vardamaan\0employee_crud>python -m pip insta

Requirement already satisfied: mysql-connector-python in c:\users\parab

n\python313\lib\site-packages (9.4.0)

[notice] A new release of pip is available: 25.2 -> 26.1.2

[notice] To update, run: python.exe -m pip install --upgrade pip

C:\-----Working-----\0_Hyd_Vardamaan\0employee_crud>python.exe -m pip i

Requirement already satisfied: pip in c:\users\parab\appdata\local\prog

e-packages (25.2)

Collecting pip

Downloading pip-26.1.2-py3-none-any.whl.metadata (4.6 kB)

Downloading pip-26.1.2-py3-none-any.whl (1.8 MB)

1.8/1.8 MB 743.2 kB/s 0:00

Installing collected packages: pip

Attempting uninstall: pip

Found existing installation: pip 25.2

Uninstalling pip-25.2:

Successfully uninstalled pip-25.2

Successfully installed pip-26.1.2

< 0 0 0

Connect

Ln 1, Col 1 Sp

C:\WINDOWS\system32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6466]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\parab>python --version  
Python 3.13.7
```

```
C:\Users\parab>python -m pip --version  
pip 26.1.2 from C:\Users\parab\AppData\Local\Programs\Python\Python31
```

```
C:\Users\parab>
```

Step 2: Create MySQL Database

Open MySQL.

CREATE DATABASE *pythoncompanydb*;
USE *pythoncompanydb*;

Create table.

**CREATE TABLE employee(
id INT PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(100),
department VARCHAR(100),
salary DECIMAL(10,2)
);**

Check table.

DESC *employee*;

Step 3: Create Database Connection

db.py

```
import mysql.connector
```

```
def get_connection():
```

```
    connection = mysql.connector.connect(
        host="localhost",
        user="root",
        password="your_password",
        database="companydb"
    )
```

```
    return connection
```

Replace: your_password with your MySQL password.

Step 4: Create CRUD Program

employee_crud.py

```
from db import get_connection
```

```
def add_employee():
```

```
    conn = get_connection()
```

```
    cursor = conn.cursor()
```

```
    name = input("Enter Name: ")
```

```
department = input("Enter Department: ")
salary = float(input("Enter Salary: "))
sql = """
INSERT INTO employee
(name,department,salary)
VALUES(%s,%s,%s)
"""

values = (name, department, salary)
cursor.execute(sql, values)
conn.commit()
print("Employee Added")
cursor.close()
conn.close()
```

```
def view_employees():
    conn = get_connection()
    cursor = conn.cursor()
    cursor.execute("SELECT * FROM employee")
    records = cursor.fetchall()
    print()
    print("Employees")
    for row in records:
        print(row)
    cursor.close()
    conn.close()
```

```
def update_employee():  
    conn = get_connection()  
    cursor = conn.cursor()  
    employee_id = int(input("Enter Employee ID: "))  
    salary = float(input("Enter New Salary: "))  
    sql = ""  
    UPDATE employee  
    SET salary=%s  
    WHERE id=%s  
    ""  
  
    values = (salary, employee_id)  
    cursor.execute(sql, values)  
    conn.commit()  
    print("Updated Successfully")  
    cursor.close()  
    conn.close()
```

```
def delete_employee():  
    conn = get_connection()  
    cursor = conn.cursor()  
    employee_id = int(input("Enter Employee ID: "))  
    sql = ""  
    DELETE FROM employee  
    WHERE id=%s  
    ""  
  
    cursor.execute(sql, (employee_id,))  
    conn.commit()  
    print("Deleted Successfully")  
    cursor.close()  
    conn.close()
```



```
def menu():
    while True:
        print()
        print("1.Add Employee")
        print("2.View Employees")
        print("3.Update Employee")
        print("4.Delete Employee")
        print("5.Exit")
        choice = input("Enter Choice: ")
        if choice == "1":
            add_employee()
        elif choice == "2":
            view_employees()
        elif choice == "3":
            update_employee()
        elif choice == "4":
            delete_employee()
        elif choice == "5":
            print("Thank You")
            break
        else:
            print("Invalid Choice")
```

menu() # calling

Step 5: Run Program

python employee_crud.py

Real-Time Flow Diagram

User



Python Menu



CRUD Function



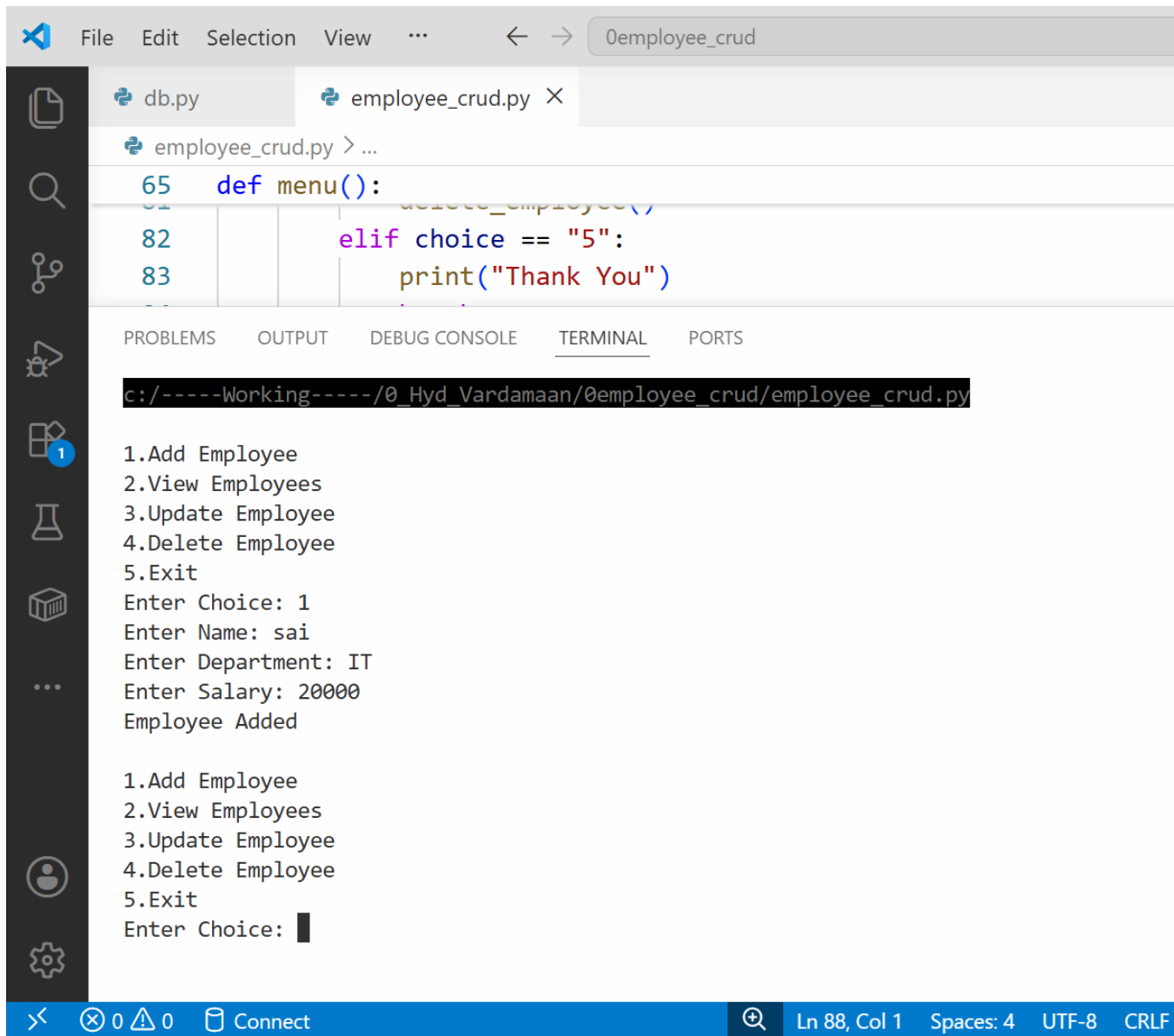
MySQL Connector



MySQL Database



Employee Table



```
File Edit Selection View ... 0employee_crud
db.py employee_crud.py X
employee_crud.py > ...
65 def menu():
82 elif choice == "5":
83 print("Thank You")
...
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
c:/-----Working-----/0_Hyd_Vardamaan/0employee_crud/employee_crud.py

1.Add Employee
2.View Employees
3.Update Employee
4.Delete Employee
5.Exit
Enter Choice: 1
Enter Name: sai
Enter Department: IT
Enter Salary: 20000
Employee Added

1.Add Employee
2.View Employees
3.Update Employee
4.Delete Employee
5.Exit
Enter Choice: █
```

1.Add Employee
2.View Employees
3.Update Employee
4.Delete Employee
5.Exit
Enter Choice: 2

Employees

(1, 'sai', 'IT', Decimal('20000.00'))

MySQL Shell

MySQL localhost:33060+ ssl pythoncompanydb SQL > show tables

```
+-----+
| Tables_in_pythoncompanydb |
+-----+
| employee                   |
+-----+
```

1 row in set (0.0020 sec)

MySQL localhost:33060+ ssl pythoncompanydb SQL > select

```
+----+-----+-----+-----+
| id | name | department | salary |
+----+-----+-----+-----+
|  1 | sai  | IT         | 20000.00 |
+----+-----+-----+-----+
```

1 row in set (0.0012 sec)

MySQL localhost:33060+ ssl pythoncompanydb SQL >

1.Add Employee

2.View Employees

3.Update Employee

4.Delete Employee

5.Exit

Enter Choice: 3

Enter Employee ID: 1

Enter New Salary: 60000

Updated Successfully

```
MySQL localhost:33060+ ssl pythoncompanydb SQL > select
+----+-----+-----+-----+
| id | name | department | salary |
+----+-----+-----+-----+
| 1  | sai  | IT         | 20000.00 |
+----+-----+-----+-----+
1 row in set (0.0012 sec)

MySQL localhost:33060+ ssl pythoncompanydb SQL > select
+----+-----+-----+-----+
| id | name | department | salary |
+----+-----+-----+-----+
| 1  | sai  | IT         | 60000.00 |
+----+-----+-----+-----+
```